

## CSE 321 Study Question Examples

- 1) Consider the Mergesort algorithm.
  - a) Give an example of an 8-element array which requires the maximum number of comparisons. Explain why.
  - b) Give an example of an 8-element array which requires the minimum number of comparisons. Explain why.

For each of the problem below:

- design a divide and conquer algorithm
  - write your algorithm in pseudocode
  - analyze the running time of your algorithm
  - analyze the space complexity of your algorithm
- 2) Given a non-decreasing array  $A[1..n]$  of integers such that consecutive numbers differ by at most 1 (that is, for every  $i = 1, 2, \dots, n - 1$ ,  $|A[i] - A[i + 1]| \leq 1$ ). Assume that  $A[1] < 0 < A[n]$  and  $A[i] = 0$  for some  $i$ . Design a divide and conquer algorithm to find an index  $i$  such that  $A[i] = 0$ .
  - 3) As part of the training of combat candidate pilots, it is aimed to measure their performance on making the right decisions in panic. For this purpose, they take off with planes with low fuel for a flight. They must request refueling from the nearest tanker aircraft before the plane lands. The one who can get the fuel from the nearest aircraft completes the test successfully. Help pilots identify the closest by designing a divide and conquer algorithm. Assume that the position of tanker aircrafts are known.
  - 4) A bottle factory produces bottles of equal mass. During a production, the weight of one of the bottles is set incorrectly. The weighing will be used to find this bottle. Design a divide-and-conquer algorithm which finds that bottle.